

Supplementary material

Physics-augmented neural networks and projection-based reduced-order models for anisotropic hyperelasticity

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This supplement documents additional material-A constitutive diagnostics—out-of-domain response, sampled finite-strain curvature, and closed-cycle work—together with technical details of the latent-space closure used by the reduced microscopic model. The cycle audit includes both learned constitutive laws and reduced microscopic models.

S1 Out-of-domain constitutive response

The available material-A fitting data occupy the engineering-strain box

$$\mathbf{e}_{\min} = (0, -0.09865, -0.16807)^T, \quad \mathbf{e}_{\max} = (0.1955, 0.01635, 0.10793)^T, \quad (\text{S1})$$

with $\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T$. The independent test set contains 400 states. An additional 350-state probe extends selected distances from the box center by factors 1.05, 1.10, 1.25, 1.50, and 2.00. Five probe labels are nonfinite following failed reference solves and are excluded with the same finite-label mask for all compared models. Their exclusion is reported, not interpreted as proof of a physical instability. Test and probe labels were not used in the selected polyconvex runs’ second-stage checkpoint selection.

Free has aggregate stress errors of 1.2643% on the test set and 5.5024% on the finite probe set. The selected polyconvex models improve the test fit but do not outperform Free on aggregate probe error. These are archived-protocol comparisons, not matched-budget architecture comparisons.

Table S1: Material A: archived test and probe constitutive errors, in percent.

Model	e_S^{test}	e_W^{test}	e_S^{probe}	Worst test	Worst probe
ICNN	0.6491	0.2169	7.0582	5.53	43.60
ICKAN	0.6551	0.2148	7.0227	5.78	45.92

All values are percentages. The last two columns are maximum per-state stress errors; the first three are aggregate array norms. Test: 400 states; probe: 345 finite reference states out of 350.

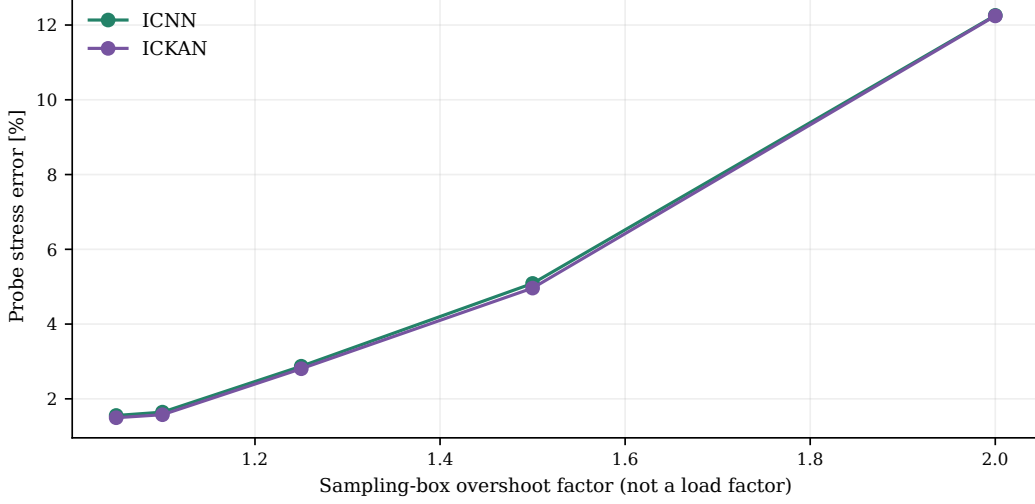


Figure S1: Stress error on increasingly distant probe rings, using only finite reference labels. Ring factors expand the sampling box; they are not coupon load multipliers. The increase to approximately 12.25% at factor 2 illustrates a remaining extrapolation limitation despite architectural admissibility.

S2 Rank-one curvature: samples and an unresolved reference

For $\mathbf{H} = \mathbf{a} \otimes \mathbf{b}$, the relevant finite-strain curvature is

$$D_{\mathbf{F}}^2 W[\mathbf{H}, \mathbf{H}] = (\delta \mathbf{e})^T \mathbf{D} \delta \mathbf{e} + \mathbf{S} : (\mathbf{H}^T \mathbf{H}), \quad \delta \mathbf{E} = \text{sym}(\mathbf{F}^T \mathbf{H}), \quad (\text{S2})$$

where $\delta \mathbf{e} = (\delta E_{11}, \delta E_{22}, 2\delta E_{12})^T$, $\mathbf{D}$ is the engineering-strain stress tangent, and $\mathbf{S}$ is the second Piola stress. Here $\mathbf{a}, \mathbf{b} \in \mathbb{R}^2$ define the rank-one direction and $\text{sym}(\mathbf{A}) = (\mathbf{A} + \mathbf{A}^T)/2$. Omitting the second term tests a different property. In particular, a positive semidefinite tangent with respect to $\mathbf{e}$ does not generally imply rank-one convexity in compression. For energy-based models, Eq. (S2) is evaluated over sampled states and unit directions; negative candidates are checked by second differences of $W(\mathbf{F} \pm h\mathbf{H})$.

Table S2: Finite rank-one curvature audit. Positive sampled values do not establish a global property.

State set	N	Free	ICNN	ICKAN
Held-out test	400	176.681	220.486	220.468
Finite-label probe	345	113.034	203.280	203.538
In-box audit (unlabelled)	12000	163.040	213.979	211.168

Minimum sampled rank-one curvature in MPa. Minima need not occur at the same state or direction. The additional in-box cloud has no independently computed FOM labels; 180 unit directions for $\mathbf{b}$ are sampled, with minimization over $\mathbf{a}$ via an eigenproblem. None of these finite samples is a global certificate.

No negative direction is found for Free in the audited test, finite-label probe, and additional in-box sets. An outward search over 1800 strain-space rays finds its first sampled negative value at expansion factor 1.55, with $J \simeq 0.780$ and curvature approximately -3.43 MPa. This is a result of that search, not the exact nearest violation. Independent energy finite differences confirm the negative curvature of the learned model.

However, the attempted FOM continuation fails numerically before reaching that candidate. No refined-continuation or global branch-tracking result is available there. Therefore this candidate establishes that the Free network lacks global rank-one convexity, but does not establish disagreement with a converged stable homogenized reference at that state. Solver nonconvergence is not evidence that the physical branch ceases to exist. The stronger claim of an FOM-confirmed false instability is left open. The ICNN and ICKAN have an analytical polyconvexity argument; their sampling audit is a check of implementation, not its replacement.

S3 Closed-cycle work of the effective stress

For a hyperelastic law, returning to the same deformation on the same reversible equilibrium branch returns the stored energy to its initial value. The net stress work must therefore vanish:

$$\mathcal{W}_{\text{cyc}} = \oint \mathbf{s} \cdot d\mathbf{e} = 0, \quad \mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T, \quad \mathbf{s} = (S_{11}, S_{22}, S_{12})^T. \quad (\text{S3})$$

Here $\mathbf{e}$ and $\mathbf{s}$ are work-conjugate strain and stress vectors. A nonzero result beyond numerical error rules out an exact potential for that stress map; a zero result on one cycle cannot establish one. The test does not assess polyconvexity.

We use a rectangle of half-width 0.001 in (E_{11}, E_{22}) , with fixed shear and center

$$\mathbf{e}_c = (0.0051260033, -0.0945507613, -0.1283720602)^T.$$

It lies inside material A's fitting box. The center was selected after observing Regression tangent asymmetry; the same cycle was subsequently applied to the frozen reduced models.

Table S3: Net work around the counterclockwise material-A cycle, in J m^{-3} .

Model	$\mathcal{W}_{\text{cyc}}$
FOM	-6.98×10^{-10}
Regression	-213.922
Free	2.34×10^{-10}
ICNN	3.19×10^{-10}
ICKAN	4.62×10^{-6}
HPROM	-1.234
HPROM-ANN	247.774
D-HPROM-ANN	54.082

128 Gauss points per edge; FOM uses 8. The ICKAN residual decreases with quadrature refinement through its spline knots.

Free, ICNN, and ICKAN obtain stress by differentiating an energy; their residuals are consistent with numerical integration and roundoff. Regression and the deployed HPROM outputs retain nonzero work under quadrature refinement. Reversing the cycle reverses its sign, and tighter equilibrium tolerances leave the intrusive residuals essentially unchanged.

The HPROMs fit equilibrium and stress cubatures separately, so their effective stresses are not guaranteed to derive from one energy. This diagnostic concerns those implementations. Among them, the largest ratio of absolute net work to accumulated absolute work along this cycle is 0.0205%. Its effect on coupon accuracy has not been demonstrated. The selected cycle establishes neither a general ranking of the models nor the frequency of such defects across the strain domain.

S4 Latent-space closure details

This section supplies the algebra omitted from Section 4.3 of the main manuscript. Let $\Phi \in \mathbb{R}^{N \times n_{\text{tot}}}$ be the retained orthonormal POD basis, let $\mathbf{S}_{\text{snap}}$ collect the centered displacement snapshots, and collect the corresponding engineering-strain inputs in $\mathbf{M} = [\mathbf{e}^1, \dots, \mathbf{e}^{N_s}]$. Their POD coefficients are

$$\mathbf{Q}_{\text{POD}} = \Phi^T \mathbf{S}_{\text{snap}}. \quad (\text{S4})$$

The input-informed construction first identifies the least-squares map from these coefficients to strain,

$$\mathbf{T}_m = \mathbf{M} \mathbf{Q}_{\text{POD}}^\dagger \in \arg \min_H \|\mathbf{H} \mathbf{Q}_{\text{POD}} - \mathbf{M}\|_F^2, \quad (\text{S5})$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. Assuming rank three, the compact factorization

$$\Phi \mathbf{T}_m^T = \mathbf{V} \Sigma_m \mathbf{Z}_m^T, \quad \mathbf{A}_m = \Sigma_m^{-1} \mathbf{Z}_m^T \quad (\text{S6})$$

defines the three-column orthonormal primary block $\mathbf{V}$ and its coordinate scaling $\mathbf{A}_m$. The secondary block $\bar{\mathbf{V}}$ is an orthonormal complement within the retained POD span, ordered by snapshot energy. With $\mathbf{q} = \mathbf{A}_m \boldsymbol{\xi}$, the deployed decoder is

$$\tilde{\mathbf{u}}(\boldsymbol{\xi}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \boldsymbol{\xi} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\boldsymbol{\xi}), \quad \boldsymbol{\xi}^s = \mathbf{T}_m \mathbf{Q}_{\text{POD}}^s. \quad (\text{S7})$$

Thus $\boldsymbol{\xi}^s \approx \boldsymbol{e}^s$ is a fitted snapshot relation, not a kinematic identity.

The displacement error separates the two approximation mechanisms. For $\mathbf{V}_{\text{tot}} = [\mathbf{V}, \overline{\mathbf{V}}]$ and projected coordinates $(\mathbf{q}^s, \bar{\mathbf{q}}^s)$,

$$\|\mathbf{u}^s - \tilde{\mathbf{u}}(\mathbf{q}^s)\|_2^2 = \|\bar{\mathbf{q}}^s - \mathcal{N}(\mathbf{q}^s)\|_2^2 + \|(\mathbf{I} - \mathbf{V}_{\text{tot}} \mathbf{V}_{\text{tot}}^T)(\mathbf{u}^s - \mathbf{u}_{\text{ref}})\|_2^2. \quad (\text{S8})$$

The first term is the closure error within the retained span; the second is the POD truncation error. Increasing network capacity cannot recover a displacement component excluded from $\mathbf{V}_{\text{tot}}$.

Finally, let $\mathbf{B}_\xi = \partial_\xi \tilde{\mathbf{u}}$, $\mathbf{R}$ be the constrained microscopic residual, and $\mathbf{K} = \partial_{\mathbf{u}} \mathbf{R}$. Differentiating the projected residual $\mathbf{r} = \mathbf{B}_\xi^T \mathbf{R}$ gives

$$\frac{\partial \mathbf{r}}{\partial \boldsymbol{\xi}} = \mathbf{B}_\xi^T \mathbf{K} \mathbf{B}_\xi + \sum_{j=1}^N R_j \frac{\partial^2 \tilde{u}_j}{\partial \boldsymbol{\xi} \partial \boldsymbol{\xi}}. \quad (\text{S9})$$

The second term is the curvature contribution of the decoder manifold. It need not vanish at reduced equilibrium because $\mathbf{B}_\xi^T \mathbf{R} = \mathbf{0}$ does not imply $\mathbf{R} = \mathbf{0}$.